Steps of the Machine Learning

- Data Collection
- EDA
    - Data Cleaning:
    - Data Analysis
    - Pre-processing
- Split the Data in to Training and Testing
- Transform
    - Encoding
    - Scaling
- Model Building
    - Model training
        - Model Evaluation
    - Prediction Pipeline

Model Building

- Training
- Prediction

Training Phase

a) Choose model based on problem
b) Hyper parameter tuning
c) Model evaluation
d) Select the best model
e) Save model as pkl file

Prediction Phase

a) Create the API to predict the new data
b) Handover the API to user (aws, or any other cloud)
c) API ready for prediction

x						y
CustID	Age	Monthly Spend	Tenure Months	Support Calls	Payment Method	Return
1001	56	79.7	22	3	Bank Transfer	Yes
1002	46	63.38	44	0	Bank Transfer	No
1003	32	38.57	25	2	Bank Transfer	No
1004	60	104.62	27	4	Credit Card	No
1005	25	27.33	42	2	Bank Transfer	Yes
1006	38	113.86	28	4	Credit Card	No
1007	56	142.01	16	0	Bank Transfer	Yes
1008	36	20.1	15	1	Electronic Check	No
1009	40	148.99	47	3	Electronic Check	No
1010	28	100.27	44	0	Credit Card	Yes

- **EDA**
  1) **Data Cleaning:**
  2) **Data Analysis**
  3) **Pre-processing**

X= Independent Variable
Y= Dependent Variable

x						y
CustID	Age	Monthly Spend	Tenure Months	Support Calls	Payment Method	Return
1001	56	79.7	22	3	Bank Transfer	Yes
1002	46	63.38	44	0	Bank Transfer	No
1003	32	38.57	25	2	Bank Transfer	No
1004	60	104.62	27	4	Credit Card	No
1005	25	27.33	42	2	Bank Transfer	Yes
1006	38	113.86	28	4	Credit Card	No
1007	56	142.01	16	0	Bank Transfer	Yes
1008	36	20.1	15	1	Electronic Check	No
1009	40	148.99	47	3	Electronic Check	No
1010	28	100.27	44	0	Credit Card	Yes

X= Independent Variable

Y= Dependent Variable

X_train

CustID	Age	Monthly Spend	Tenure Months	Support Calls	Payment Method
1001	56	79.7	22	3	Bank Transfer
1002	46	63.38	44	0	Bank Transfer
1003	32	38.57	25	2	Bank Transfer
1004	60	104.62	27	4	Credit Card
1005	25	27.33	42	2	Bank Transfer
1006	38	113.86	28	4	Credit Card
1007	56	142.01	16	0	Bank Transfer

y_train_true

Return
Yes
No
No
No
Yes
No
Yes

X_test

CustID	Age	Monthly Spend	Tenure Months	Support Calls	Payment Method
1008	36	20.1	15	1	Electronic Check
1009	40	148.99	47	3	Electronic Check
1010	28	100.27	44	0	Credit Card

y_test_true

Retrun
No
No
Yes

- Split the Data in to Training and Testing
  - Training 70%
  - Testing 30%

```
X_train_trans = trans_Obj.fit_transform(X_train)
X_test_trans  = trans_Obj.transform(X_test)
```

After Encoding and Scaling

X_Train_trans						
Age	Monthly Spend	Tenure Months	Support Calls	Bank Transfer	Credit Card	Electronic Check
0.91	-0.04	-0.75	0.55	1	0	0
0.1	-0.47	1.56	-1.38	1	0	0
-1.02	-1.11	-0.44	-0.09	1	0	0
1.23	0.61	-0.23	1.2	0	1	0
-1.58	-1.4	1.35	-0.09	1	0	0
-0.54	0.85	-0.12	1.2	0	1	0
0.91	1.58	-1.38	-1.38	1	0	0

X_test_trans						
Age	Monthly Spend	Tenure Months	Support Calls	Bank Transfer	Credit Card	Electronic Check
0.27	-1.31	-1.41	-0.27	0	0	1
1.07	1.11	0.81	1.34	0	0	1
-1.34	0.2	0.6	-1.07	0	1	0

Training (Training to be done on X_train Data)

x_train_trans

Age	Monthly Spend	Tenure Months	Support Calls	Bank Transfer	Credit Card	Electronic Check
0.91	-0.04	-0.75	0.55	1	0	0
0.1	-0.47	1.56	-1.38	1	0	0
-1.02	-1.11	-0.44	-0.09	1	0	0
1.23	0.61	-0.23	1.2	0	1	0
-1.58	-1.4	1.35	-0.09	1	0	0
-0.54	0.85	-0.12	1.2	0	1	0
0.91	1.58	-1.38	-1.38	1	0	0




```
Training (Training to be done on X_train Data)

from sklearn.linear_model import LinearRegression
LinerModel = LinearRegression()
LinerModel.fit(X_train_trans,y_train_true)
```

```
Save model to the file
joblib.dump(LinerModel,"Airtifacts/pklModels/Model.pkl")
```

x_train_trans

Age	Monthly Spend	Tenure Months	Support Calls	Bank Transfer	Credit Card	Electronic Check
0.91	-0.04	-0.75	0.55	1	0	0
0.1	-0.47	1.56	-1.38	1	0	0
-1.02	-1.11	-0.44	-0.09	1	0	0
1.23	0.61	-0.23	1.2	0	1	0
-1.58	-1.4	1.35	-0.09	1	0	0
-0.54	0.85	-0.12	1.2	0	1	0
0.91	1.58	-1.38	-1.38	1	0	0

```
y_train_pred = LinerModel_01.predict(X_train_trans)
```

Model.pkl

y_train_pred

Return
1
1
1
0
1
0
1

X_train_trans

Age	Monthly Spend	Tenure Months	Support Calls	Bank Transfer	Credit Card	Electronic Check
0.91	-0.04	-0.75	0.55	1	0	0
0.1	-0.47	1.56	-1.38	1	0	0
-1.02	-1.11	-0.44	-0.09	1	0	0
1.23	0.61	-0.23	1.2	0	1	0
-1.58	-1.4	1.35	-0.09	1	0	0
-0.54	0.85	-0.12	1.2	0	1	0
0.91	1.58	-1.38	-1.38	1	0	0

y train_true

Return
1
0
0
0
1
0
1

y_train_pred

Return
1
1
1
0
1
0
1

x_test_trans

Age	Monthly Spend	Tenure Months	Support Calls	Bank Transfer	Credit Card	Electronic Check
0.27	-1.31	-1.41	-0.27	0	0	1
1.07	1.11	0.81	1.34	0	0	1
-1.34	0.2	0.6	-1.07	0	1	0

```
# Extract y_test_pred
y_test_pred = LinerModel_01.predict(X_test_trans)
y_test_pred
```



x_train_trans

Age	Monthly Spend	Tenure Months	Support Calls	Bank Transfer	Credit Card	Electronic Check
0.91	-0.04	-0.75	0.55	1	0	0
0.1	-0.47	1.56	-1.38	1	0	0
-1.02	-1.11	-0.44	-0.09	1	0	0
1.23	0.61	-0.23	1.2	0	1	0
-1.58	-1.4	1.35	-0.09	1	0	0
-0.54	0.85	-0.12	1.2	0	1	0
0.91	1.58	-1.38	-1.38	1	0	0

y_train_true

Return
1
0
0
0
1
0
1

y_train_pred

Return
1
1
1
0
1
0
1

x_test_trans

Age	Monthly Spend	Tenure Months	Support Calls	Bank Transfer	Credit Card	Electronic Check
0.27	-1.31	-1.41	-0.27	0	0	1
1.07	1.11	0.81	1.34	0	0	1
-1.34	0.2	0.6	-1.07	0	1	0

y_test_true

Return
0
0
1

y_test_pred

Return
1
0
1

Evaluation Phase

- Training Data Evaluation
- Testing Data Evaluation

Training

Data Evaluation

y_train_true

Return
1
0
0
0
1
0
1

y_train_pred

Return
1
1
1
0
1
0
1

Testing

Data Evaluation

y_test_true

Return
0
0
1

y_test_pred

Return
1
0
1

Data Collection

EDA

1) **Data Cleaning:**
2) **Data Analysis**
3) **Advance Pre-processing**
    - Handel Imbalance Data
    - Feature Selection
    - Dimension Reduction
4) **Encoding**
5) **Split the data Training and Testing**
    - **Training Data 80%,**
    - **Testing Data20%**

- Scaling
- Model Building
    - **Model training**
    - **Model Evaluation**
    - **Prediction Pipeline**